Portfolio Specification

Data Science & Machine Learning 1 – Software Development Year 4



Author: Mark Lambert
Lecturer: Ben O'Shaughnessy
Student ID: C00192497
Submission Date: 9th October 2026

1. Overview and Background

In Semester 1 of Year 3 we were taught Python, in parallel with coursework I built a Last.Fm scrobbling application that connects with the Last.Fm API to track my music listening in order to step away from Spotify and subscription streaming services. I deployed the application to pythonanywhere, as shown in our Python Web Dev classes.

A year has passed, and as such I now have a year's worth of data that would allow me to personally connect with my portfolio. Figure 1 shows just under 18,000 entries.



Figure 1: 17,927 data points tracking my music listening over the last 365 days

1.1 Dataset

- **Source:** Data can be pulled from the Last.Fm API. The endpoint of focus for this portfolio is `user.getRecentTracks`, which returns a users own scrobble history and `user.getLovedTracks`, which will be joined onto the scrobble data as a binary field.
- **Size:** ~18,000 timestamped scrobbles collected personally via API key
- **Fields:** artist, album, song, timestamp, is_loved | Processing scrobbles into daily and monthly counts for modelling

The Data as a raw format once fetched from the API can be seen in Figure 2.

[4]:

	artist	streamable	image	mbid	album	name	@attr	url	dat
0	{'mbid': 'be407b02-f3e6-4ed5-9489-f8e5f0ab36dc...	0	[{'size': 'small', '#text': 'https://lastfm-im...	253969ba-738f-351f-9109-e596f7fd35c5	{'mbid': '0ed51cc9-4840-45ab-8b6d-f71dd9cbad15...	You Don't Understand Me	{'nowplaying': 'true'}	https://www.last.fm/music/The+Raconteurs/_/You...	Nal
1	{'mbid': 'be407b02-f3e6-4ed5-9489-f8e5f0ab36dc...	0	[{'size': 'small', '#text': 'https://lastfm-im...	06795db7-95ef-3a46-8de0-4940150664df	{'mbid': '0ed51cc9-4840-45ab-8b6d-f71dd9cbad15...	Salute Your Solution	NaN	https://www.last.fm/music/The+Raconteurs/_/Sal...	{'uts '1791298940 '#text': '0 Oct 2026, 1.
2	{'mbid': 'be407b02-f3e6-4ed5-9489-f8e5f0ab36dc...	0	[{'size': 'small', '#text': 'https://lastfm-im...	0955e452-52f6-3c5e-9124-605863694b77	{'mbid': '0ed51cc9-4840-45ab-8b6d-f71dd9cbad15...	Consoler of the Lonely	NaN	https://www.last.fm/music/The+Raconteurs/_/Con...	{'uts '1791298760 '#text': '0 Oct 2026, 1.

Figure 2: Raw data response from API

The details regarding each artist, album and date are all hashed with an embedded #text attribute, which is detailed in the Unofficial Last.Fm API docs [2]. The goal is to filter the data to a readable format, with dates, timestamps and the date + time in a readable format.

2. Technologies

- **Language:** Python
- **Data Science & ML libraries:** pandas, scikit-learn
- **Visualisation:** matplotlib
- **Environment:** Jupyter Notebook
- **Other Libraries:** Python's requests module for connecting with Last.Fm API
- **External Tools:** Last.Fm API for data collection
- **Version Control:** GitHub

3. Questions to Explore

- **Linear Regression:** Look at the daily trend of scrobbles over the last year

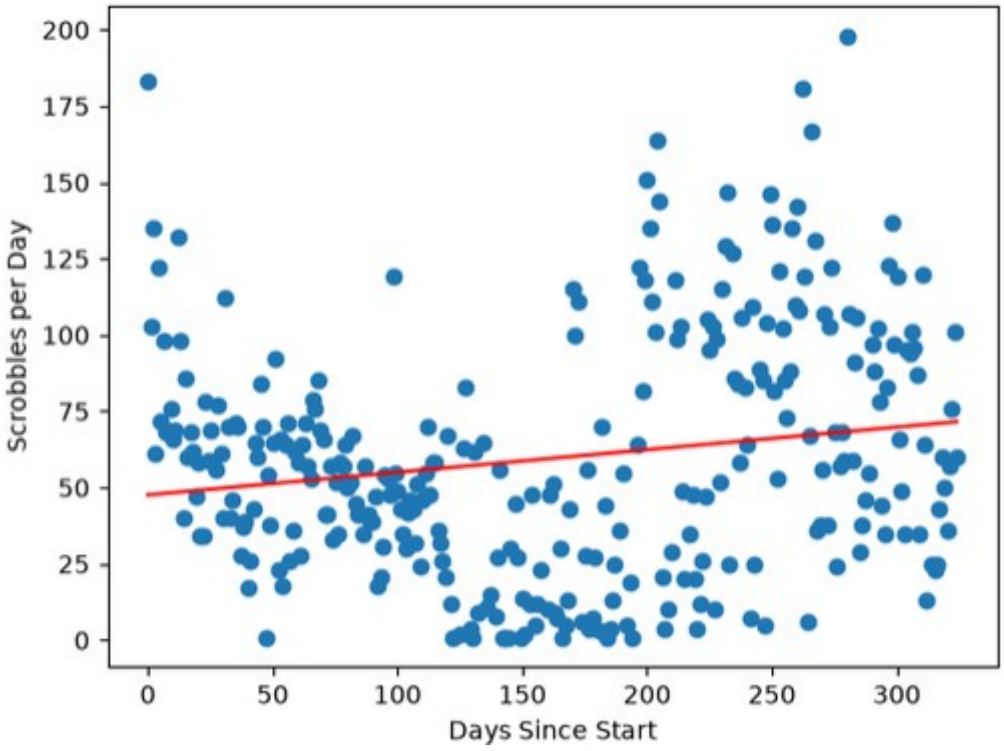


Figure 3: Linear Regression model ran on the daily scrobble data

- **TBD**

References

[1] Last.Fm (n. d.) *User.getRecentTracks* [API documentation] Available at: https://www.last.fm/api/show/user.getRecentTracks (Accessed: 05th October)

[2] Last.Fm API Docs (n.d.) *User.getRecentTracks* [Unoficcial Last.Fm API docs] Available At: https://lastfm-docs.github.io/api-docs/user/getRecentTracks/ (Accessed: 05th October 2026)

[3] Carlos Medina (2021) *Using the Last Fm API to extract data and analyzing it with Python* [Hemisferios] Available At: https://medium.com/hemisferios/using-the-last-fm-api-to-extract-data-and-analyzing-it-with-python-part-1-ffe96f812455 (Accessed: 05th October 2026)

[4] Mark Lambert (2012) *Personal LastFm Profile – Source of Scrobbling data* [LastFm] Available at: https://www.last.fm/user/MilesOver666